

Blocks

Set Wheel Speed

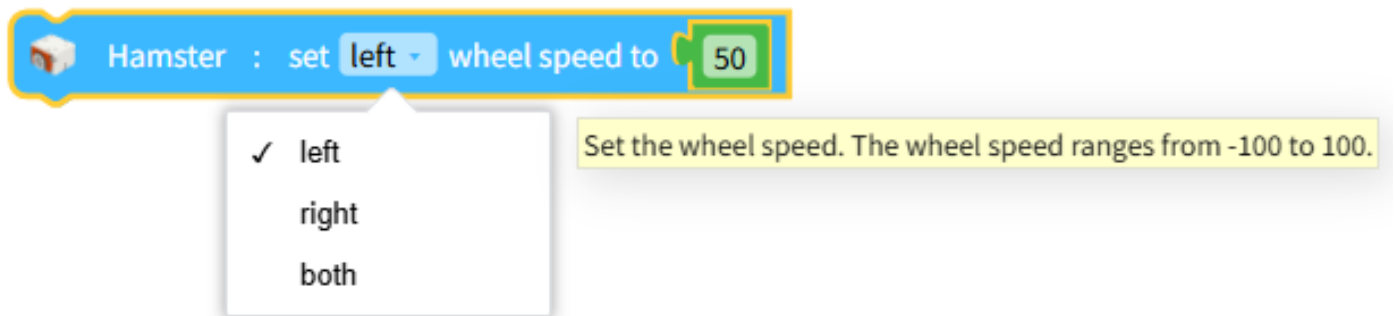
Sets the Hamster's wheel speed.

If the wheel speed is positive, it rotates forward, and if the wheel speed is negative, it rotates backward.

For example, if the wheel speed is 100, it rotates forward at speed 100,

and if the wheel speed is -100, it rotates backward at speed 100.

Once you set the wheel speed, the Hamster will keep moving at that speed until you set the wheel speed again.



Dropdown Options and Input Values

Name	Type	Description	Range / Type
wheel	Dropdown Option	Wheel type	left, right, both
velocity	Input Value	Wheel speed	Integer -100 ~ 100, 0: stop

JavaScript Code

```
// Set Hamster left wheel speed to 50
$('Hamster*0:wheel.speed.left').d = __getSpeed('Hamster*0', 50); // (robot,
speed)

// Set Hamster right wheel speed to 30
$('Hamster*0:wheel.speed.right').d = __getSpeed('Hamster*0', 30); // (robot,
speed)

// Set Hamster both wheel speeds to -10
```

```
$( 'Hamster*0:wheel.speed.left' ).d = __getSpeed( 'Hamster*0', -10); // (robot, speed)
$( 'Hamster*0:wheel.speed.right' ).d = __getSpeed( 'Hamster*0', -10); // (robot, speed)
```

Python Code

```
# Set Hamster left wheel speed to 50
__( 'Hamster*0:wheel.speed.left' ).d = __getSpeed( 'Hamster*0', 50) # (robot, speed)

# Set Hamster right wheel speed to 30
__( 'Hamster*0:wheel.speed.right' ).d = __getSpeed( 'Hamster*0', 30) # (robot, speed)

# Set Hamster both wheel speeds to -10
__( 'Hamster*0:wheel.speed.left' ).d = __getSpeed( 'Hamster*0', -10) # (robot, speed)
__( 'Hamster*0:wheel.speed.right' ).d = __getSpeed( 'Hamster*0', -10) # (robot, speed)
```

Move by Time

Sets the time the Hamster will move.

If wheel speed is not set, it will not move.

If you check “wait”, it waits until the move is completed.

If you check “wait”, it can only be used inside an async function.



✓ seconds
milliseconds

Set the time for Hamster to move.
If the wheel speed is not set, it won't move.
If wait is checked, it will wait until move completes.

Dropdown Options and Input Values

Name	Type	Description	Range / Type
time	Input Value	Time value	Real number ≥ 0
unit	Dropdown Option	Time unit	seconds, milliseconds

If you set the option to milliseconds, the input will be the value of time divided by 1000.

JavaScript Code

```
// Move 5 seconds | Wait 0
await __stopAfterDelay('Hamster*0', 5, true); // (robot, time, wait_w)

// Move 1 millisecond | Wait X
__stopAfterDelay('Hamster*0', 0.001, false); // (robot, time, wait_w)
```

Python Code

```
# Move 5 seconds | Wait 0
await __stopAfterDelay('Hamster*0', 5, True) # (robot, time, wait_w)

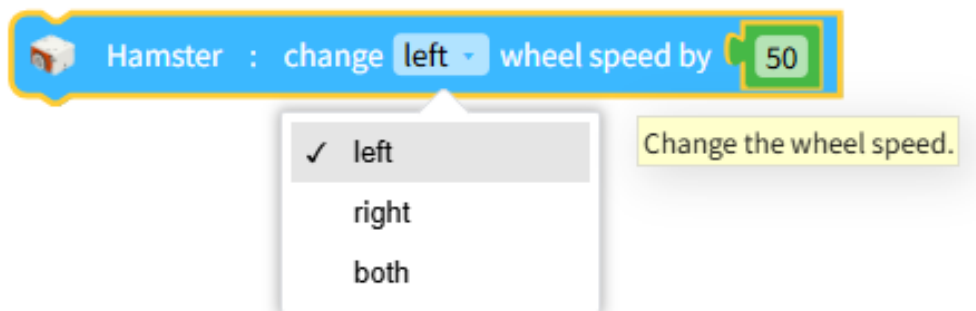
# Move 1 millisecond | Wait X
__stopAfterDelay('Hamster*0', 0.001, False) # (robot, time, wait_w)
```

Change Wheel Speed

Changes the Hamster's wheel speed.

The new wheel speed becomes the current wheel speed plus the input value.

The new wheel speed is clamped to the range -100 ~ 100.



Dropdown Options and Input Values

Name	Type	Description	Range / Type
wheel	Dropdown Option	Wheel type	left, right, both
velocity	Input Value	Value to add to current speed	Integer -200 ~ 200, 0: stop

JavaScript Code

```
// Change Hamster left wheel speed by +50
$('Hamster*0:wheel.speed.left').d = $('Hamster*0:wheel.speed.left').d +
__getSpeed('Hamster*0', 50);

// Change Hamster right wheel speed by +10
$('Hamster*0:wheel.speed.right').d = $('Hamster*0:wheel.speed.right').d +
__getSpeed('Hamster*0', 10);

// Change Hamster both wheel speeds by -20
$('Hamster*0:wheel.speed.left').d = $('Hamster*0:wheel.speed.left').d +
__getSpeed('Hamster*0', (-20));
$('Hamster*0:wheel.speed.right').d = $('Hamster*0:wheel.speed.right').d +
__getSpeed('Hamster*0', (-20));
```

Python Code

```
# Change Hamster left wheel speed by +50
__('Hamster*0:wheel.speed.left').d = __('Hamster*0:wheel.speed.left').d +
__getSpeed('Hamster*0', 50)

# Change Hamster right wheel speed by +10
__('Hamster*0:wheel.speed.right').d = __('Hamster*0:wheel.speed.right').d +
__getSpeed('Hamster*0', 10)

# Change Hamster both wheel speeds by -20
```

```

__( 'Hamster*0:wheel.speed.left' ).d = __( 'Hamster*0:wheel.speed.left' ).d +
__getSpeed( 'Hamster*0', ( -20 ) )
__( 'Hamster*0:wheel.speed.right' ).d = __( 'Hamster*0:wheel.speed.right' ).d +
__getSpeed( 'Hamster*0', ( -20 ) )

```

Stop

Stops the Hamster's movement.

Both wheel speeds are reset to 0.



Stop moving.

JavaScript Code

```

__stopMove( 'Hamster*0' );

```

Python Code

```

__stopMove( 'Hamster*0' )

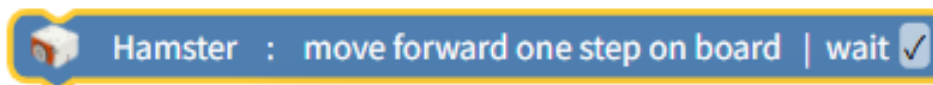
```

Move One Tile Forward on the Board

Moves the Hamster forward by one tile on the board.

If you check "wait", it waits until the move is completed.

If you check "wait", it can only be used inside an async function.



Move Hamster one step at a time on the board as specified.

JavaScript Code

```
// Move one tile forward | Wait 0
await __grid_move_forward('Hamster*0', true); // (robot, wait_w)

// Move one tile forward | Wait X
__grid_move_forward('Hamster*0', false); // (robot, wait_w)
```

Python Code

```
# Move one tile forward | Wait 0
await __grid_move_forward('Hamster*0', True) # (robot, wait_w)

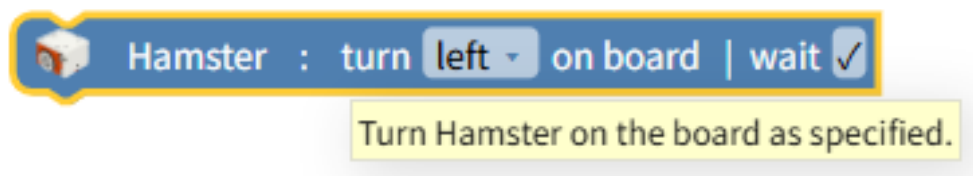
# Move one tile forward | Wait X
__grid_move_forward('Hamster*0', False) # (robot, wait_w)
```

Turn Once on the Board

Rotates the Hamster 90 degrees on the board in the given direction.

If you check “wait”, it waits until the move is completed.

If you check “wait”, it can only be used inside an async function.



Dropdown Options and Input Values

Name	Type	Description	Range / Type
direction	Dropdown Option	Turn direction	left, right

JavaScript Code

```
// Turn left once on the board | Wait 0
await __grid_turn_left('Hamster*0', true); // (robot, wait_w)

// Turn left once on the board | Wait X
__grid_turn_left('Hamster*0', false); // (robot, wait_w)

// Turn right once on the board | Wait 0
await __grid_turn_right('Hamster*0', true); // (robot, wait_w)

// Turn right once on the board | Wait X
__grid_turn_right('Hamster*0', false); // (robot, wait_w)
```

Python Code

```
# Turn left once on the board | Wait 0
await __grid_turn_left('Hamster*0', True) # (robot, wait_w)

# Turn left once on the board | Wait X
__grid_turn_left('Hamster*0', False) # (robot, wait_w)

# Turn right once on the board | Wait 0
await __grid_turn_right('Hamster*0', True) # (robot, wait_w)

# Turn right once on the board | Wait X
__grid_turn_right('Hamster*0', False) # (robot, wait_w)
```

Follow a Line Using Sensors

The Hamster follows a specific line using the floor sensors.

Hamster : trace **white** line with **left** floor sensor

Hamster : trace **black** line with **left** floor sensor

✓ left
right
center

Use the floor sensor for Hamster to trace the line of a specific color.

Dropdown Options and Input Values

Name	Type	Description	Range / Type
direction	Dropdown Option	Floor sensor position	left, right, center
color	Dropdown Option	Line color	black, white

JavaScript Code

```
$('#Hamster*0:wheel.trace.mode').d = 1; // Follow black line with left floor sensor
$('#Hamster*0:wheel.trace.mode').d = 2; // Follow black line with right floor sensor
$('#Hamster*0:wheel.trace.mode').d = 3; // Follow black line with center floor sensor
$('#Hamster*0:wheel.trace.mode').d = 8; // Follow white line with left floor sensor
$('#Hamster*0:wheel.trace.mode').d = 9; // Follow white line with right floor sensor
$('#Hamster*0:wheel.trace.mode').d = 10; // Follow white line with center floor sensor
```

Python Code

```
__('Hamster*0:wheel.trace.mode').d = 1 # Follow black line with left floor sensor
__('Hamster*0:wheel.trace.mode').d = 2 # Follow black line with right floor sensor
```

```

__('Hamster*0:wheel.trace.mode').d = 3 # Follow black line with center floor
sensor
__('Hamster*0:wheel.trace.mode').d = 8 # Follow white line with left floor sensor
__('Hamster*0:wheel.trace.mode').d = 9 # Follow white line with right floor
sensor
__('Hamster*0:wheel.trace.mode').d = 10 # Follow white line with center floor
sensor

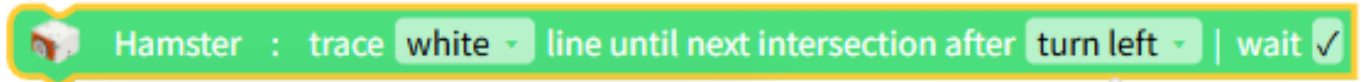
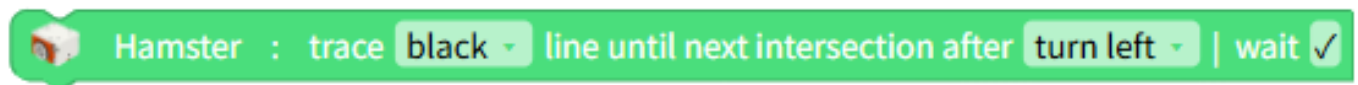
```

Turn, Then Follow Line & Stop at the Next Intersection

The Hamster turns in the specified direction, then moves along the line until it reaches the next intersection.

If you check “wait”, it waits until the move is completed.

If you check “wait”, it can only be used inside an async function.



Turn to the assigned direction and move until Hamster reaches the next intersection. If wait is checked, it will wait until trace completes.

☒ turn left
☐ turn right
☐ move forward
☐ u-turn

Dropdown Options and Input Values

Name	Type	Description	Range / Type
direction	Dropdown Option	Move direction	turn left, turn right, move forward, u-turn
color	Dropdown Option	Line color	black, white

JavaScript Code

```
// Turn left, follow black line, stop at next intersection | Wait 0
$('Hamster*0:wheel.trace.mode').d = 4;
await $('Hamster*0:wheel.trace.!mode').w();

// Turn right, follow black line, stop at next intersection | Wait X
$('Hamster*0:wheel.trace.mode').d = 5;

// Go forward, follow black line, stop at next intersection | Wait 0
$('Hamster*0:wheel.trace.mode').d = 6;
await $('Hamster*0:wheel.trace.!mode').w();

// U-turn, follow black line, stop at next intersection | Wait X
$('Hamster*0:wheel.trace.mode').d = 7;

// Turn left, follow white line, stop at next intersection | Wait 0
$('Hamster*0:wheel.trace.mode').d = 11;
await $('Hamster*0:wheel.trace.!mode').w();

// Turn right, follow white line, stop at next intersection | Wait X
$('Hamster*0:wheel.trace.mode').d = 12;

// Go forward, follow white line, stop at next intersection | Wait 0
$('Hamster*0:wheel.trace.mode').d = 13;
await $('Hamster*0:wheel.trace.!mode').w();

// U-turn, follow white line, stop at next intersection | Wait X
$('Hamster*0:wheel.trace.mode').d = 14;
```

Python Code

```
# Turn left, follow black line, stop at next intersection | Wait 0
__('Hamster*0:wheel.trace.mode').d = 4
```

```
await __('Hamster*0:wheel.trace.!mode').w()

# Turn right, follow black line, stop at next intersection | Wait X
__('Hamster*0:wheel.trace.mode').d = 5

# Go forward, follow black line, stop at next intersection | Wait 0
__('Hamster*0:wheel.trace.mode').d = 6
await __('Hamster*0:wheel.trace.!mode').w()

# U-turn, follow black line, stop at next intersection | Wait X
__('Hamster*0:wheel.trace.mode').d = 7

# Turn left, follow white line, stop at next intersection | Wait 0
__('Hamster*0:wheel.trace.mode').d = 11
await __('Hamster*0:wheel.trace.!mode').w()

# Turn right, follow white line, stop at next intersection | Wait X
__('Hamster*0:wheel.trace.mode').d = 12

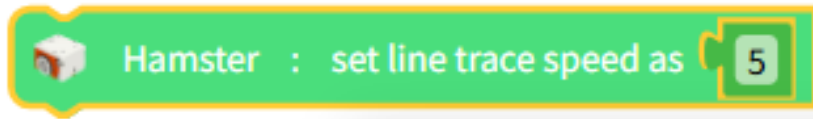
# Go forward, follow white line, stop at next intersection | Wait 0
__('Hamster*0:wheel.trace.mode').d = 13
await __('Hamster*0:wheel.trace.!mode').w()

# U-turn, follow white line, stop at next intersection | Wait X
__('Hamster*0:wheel.trace.mode').d = 14
```

Set Line Tracing Speed

Sets the Hamster's line tracing speed.

Speed range is 1 ~ 10.



Set the speed of Hamster' tracing the line.
The line trace speed ranges from 1 to 10.

Dropdown Options and Input Values

Name	Type	Description	Range / Type
velocity	Input Value	line tracing speed	Integer 1 ~ 10

JavaScript Code

```
$('#Hamster*0:wheel.trace.speed').d = 5; // Set line tracing speed to 5
```

Python Code

```
__('Hamster*0:wheel.trace.speed').d = 5 # Set line tracing speed to 5
```

Stop Line Tracing

Stops the Hamster's line tracing feature.



Terminate the line tracing function.

JavaScript Code

```
$('#Hamster*0:wheel.trace.mode').d = 0; // Stop line tracing
```

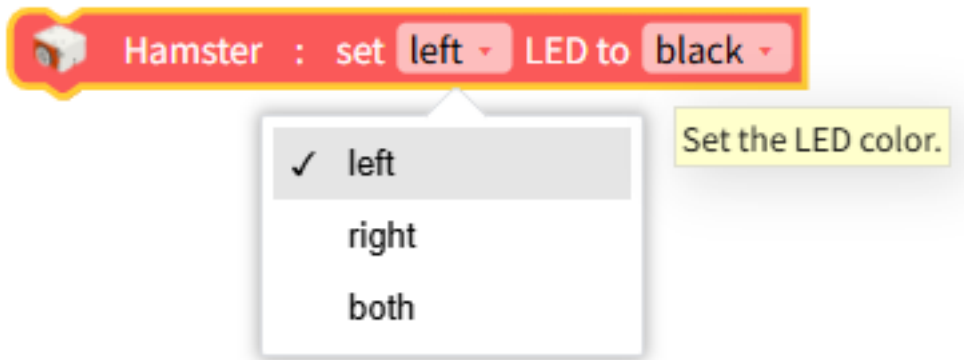
Python Code

```
__('Hamster*0:wheel.trace.mode').d = 0 # Stop line tracing
```

Set LED Color

Sets the Hamster's LED color.

You can set the left, right, or both LEDs.



Dropdown Options and Input Values

Name	Type	Description	Range / Type
direction	Dropdown Option	LED side	left, right, both
color	Dropdown Option	Color	black(0), blue(1), green(2), cyan(3), red(4), magenta(5), yellow(6), white(7)

JavaScript Code

```
// Set left LED color to red (4)
$('Hamster*0:led.left').d = 4;

// Set right LED color to green (2)
$('Hamster*0:led.right').d = 2;

// Set both LED colors to blue (1)
```

```
$( 'Hamster*0:led.left' ).d = 1;
$( 'Hamster*0:led.right' ).d = 1;
```

Python Code

```
# Set left LED color to red (4)
__('Hamster*0:led.left').d = 4

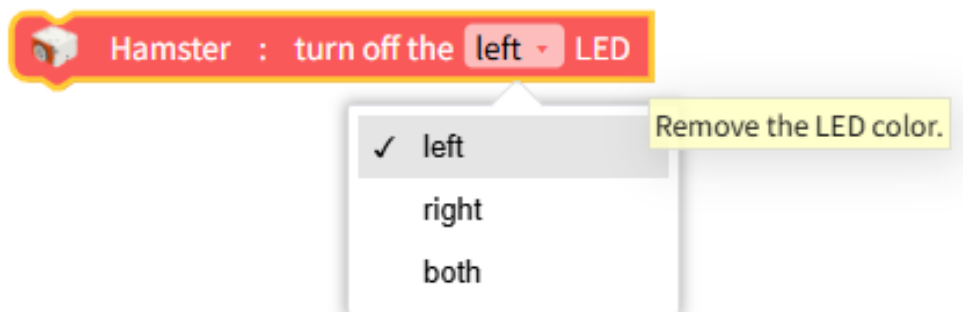
# Set right LED color to green (2)
__('Hamster*0:led.right').d = 2

# Set both LED colors to blue (1)
__('Hamster*0:led.left').d = 1
__('Hamster*0:led.right').d = 1
```

Turn Off LED

Turns off the Hamster's LED color.

You can turn off the left, right, or both LEDs.



Dropdown Options and Input Values

Name	Type	Description	Range / Type
direction	Dropdown Option	LED side	left, right, both

JavaScript Code

```
// Turn off left LED
$('Hamster*0:led.left').d = 0;

// Turn off right LED
$('Hamster*0:led.right').d = 0;

// Turn off both LEDs
$('Hamster*0:led.left').d = 0;
$('Hamster*0:led.right').d = 0;
```

Python Code

```
# Turn off left LED
__('Hamster*0:led.left').d = 0

# Turn off right LED
__('Hamster*0:led.right').d = 0

# Turn off both LEDs
__('Hamster*0:led.left').d = 0
__('Hamster*0:led.right').d = 0
```

Set Buzzer Frequency

Sets the Hamster's buzzer sound to the specified frequency.

The range of achievable frequencies is from 1.0hz to 6553.5hz.

If a value outside the range is entered, the buzzer sound will not be produced.



Set the buzzer sound with an assigned frequency.
The range of achievable frequencies is from 1.0hz to 6553.5hz.
If a value outside the range is entered,
the buzzer sound will not be produced.

Dropdown Options and Input Values

Name	Type	Description	Range / Type
sound	Input Value	Buzzer frequency	1.0 ~ 6553.5 (hz)

JavaScript Code

```
// Set buzzer frequency to 1000hz  
$('Hamster*0:sound.buzz').d = 1000;
```

Python Code

```
# Set buzzer frequency to 1000hz  
__('Hamster*0:sound.buzz').d = 1000
```

Play a Note

Plays the specified musical note on the Hamster.

Hamster : play **Do** **4** note

✓ Do
Do#(Re ♭)
Re
Re#(Mi ♭)
Mi
Fa
Fa#(Sol ♭)
Sol
Sol#(La ♭)
La

Play a specific note.

Dropdown Options and Input Values

Name	Type	Description	Range / Type
note	Dropdown Option	Note	Do, Do#, Re, Re#, Mi, Fa, Fa#, So, So#, La, La#, Si
octave	Dropdown Option	Octave	1 ~ 7

JavaScript Code

```
// Play octave 1 Do
$('Hamster*0:sound.note').d = 4;

// Play octave 1 Re
```

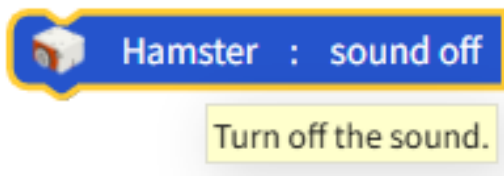
```
$( 'Hamster*0:sound.note' ).d = 6;  
  
// Play octave 2 Do  
$( 'Hamster*0:sound.note' ).d = 16;  
  
// Play octave 7 Si  
$( 'Hamster*0:sound.note' ).d = 87;
```

Python Code

```
# Play octave 1 Do  
__ ( 'Hamster*0:sound.note' ).d = 4  
  
# Play octave 1 Re  
__ ( 'Hamster*0:sound.note' ).d = 6  
  
# Play octave 2 Do  
__ ( 'Hamster*0:sound.note' ).d = 16  
  
# Play octave 7 Si  
__ ( 'Hamster*0:sound.note' ).d = 87
```

Stop Sound

Turns off the Hamster's sound.



JavaScript Code

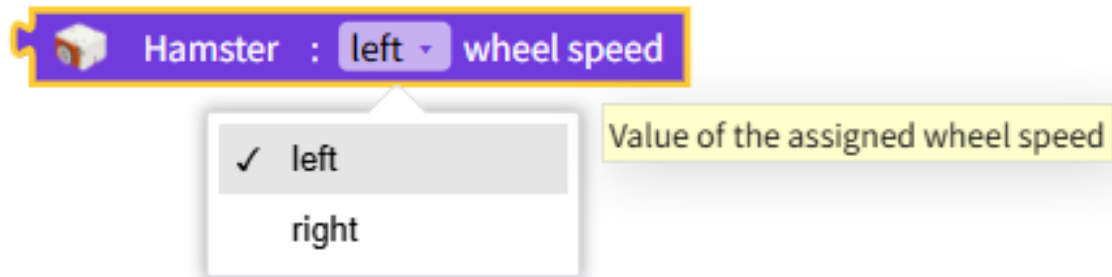
```
// Stop Hamster sound  
__stopSound( 'Hamster*0' );
```

Python Code

```
# Stop Hamster sound
__stopSound('Hamster*0')
```

Wheel Speed Value

Returns the specified Hamster wheel speed value.



Dropdown Options and Input Values

Name	Type	Description	Range / Type
direction	Dropdown Option	Direction	left, right

JavaScript Code

```
// Left wheel speed
__getSpeedInput('Hamster*0', $('Hamster*0:wheel.speed.left').d);

// Right wheel speed
__getSpeedInput('Hamster*0', $('Hamster*0:wheel.speed.right').d);
```

Python Code

```
# Left wheel speed
__getSpeedInput('Hamster*0', __('Hamster*0:wheel.speed.left').d)
```

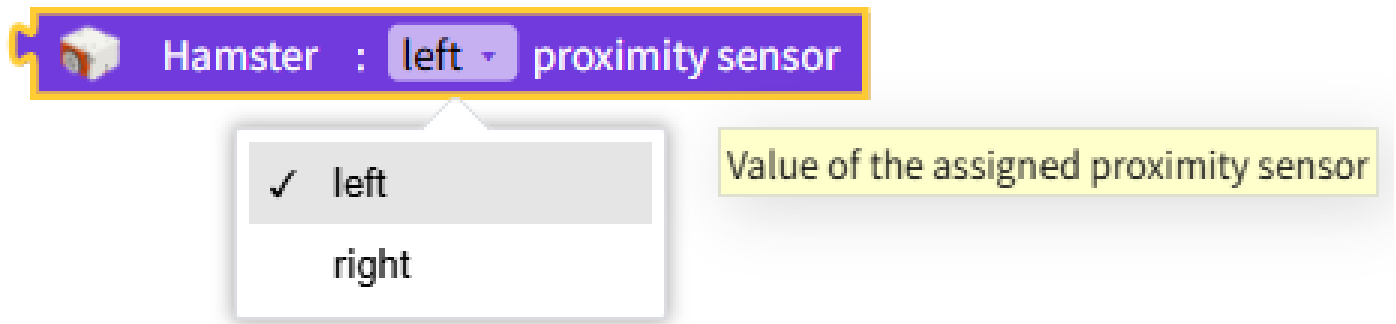
```
# Right wheel speed
__getSpeedInput('Hamster*0', __('Hamster*0:wheel.speed.right').d)
```

Proximity Sensor Value

Returns the specified Hamster proximity sensor value.

The closer the object is, the larger the value becomes.

The range is 0 to 100.



Dropdown Options and Input Values

Name	Type	Description	Range / Type
direction	Dropdown Option	Direction	left, right

JavaScript Code

```
// Left proximity sensor value
$('Hamster*0:proximity.left').d;

// Right proximity sensor value
$('Hamster*0:proximity.right').d;
```

Python Code

```
# Left proximity sensor value
__('Hamster*0:proximity.left').d
```

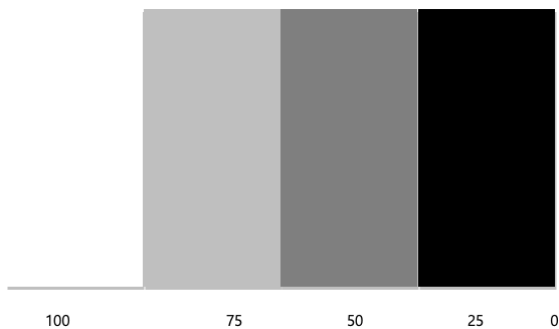
```
# Right proximity sensor value
__('Hamster*0:proximity.right').d
```

Floor Sensor Value

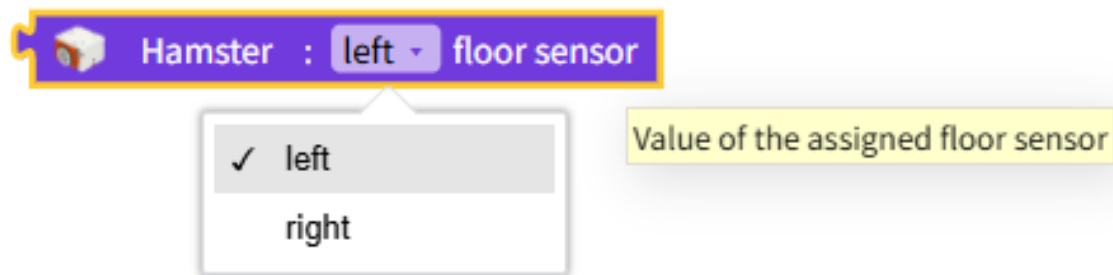
Returns the specified Hamster floor sensor value.

Black represents the minimum value, and white represents the maximum value.

The range is 0 to 100.



- White is set to 100 based on A4 paper for printers.



Dropdown Options and Input Values

Name	Type	Description	Range / Type
direction	Dropdown Option	Direction	left, right

JavaScript Code

```
// Left floor sensor value
$('Hamster*0:floor.left').d;
```

```
// Right floor sensor value
$('Hamster*0:floor.right').d;
```

Python Code

```
# Left floor sensor value
__('Hamster*0:floor.left').d

# Right floor sensor value
__('Hamster*0:floor.right').d
```

Acceleration Value

Returns the Hamster acceleration value for a specific axis.

The accelerometer is a sensor that measures how quickly Hamster moves or tilts.

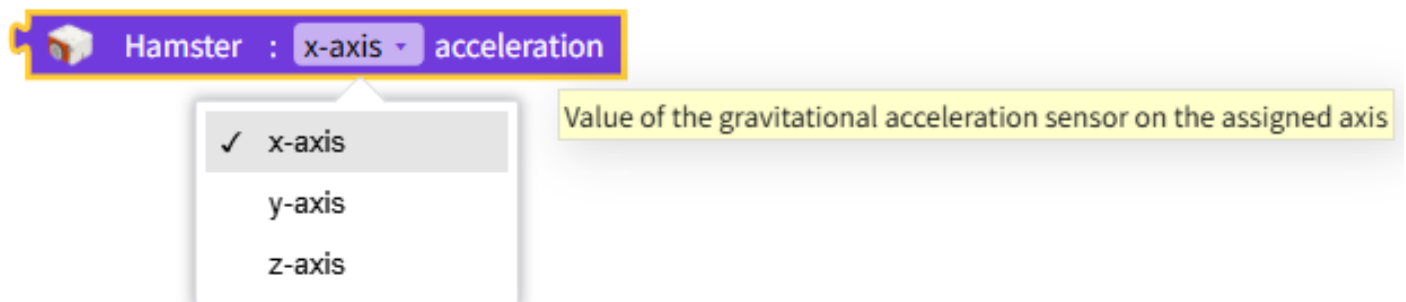
The range is -32768 to +32767.

Since Earth's gravity (1g) acts downwards, the Earth's gravity value is Z = -16384.

* X-axis: Changes when tilting the robot forward or backward.

* Y-axis: Changes when tilting the robot left or right.

* Z-axis: Changes when flipping the robot up or down.



Dropdown Options and Input Values

Name	Type	Description	Range / Type
axis	Dropdown Option	Axis	x, y, z

JavaScript Code

```
// Acceleration value on x-axis
$('Hamster*0:acceleration.x').d;

// Acceleration value on y-axis
$('Hamster*0:acceleration.y').d;

// Acceleration value on z-axis
$('Hamster*0:acceleration.z').d;
```

Python Code

```
# Acceleration value on x-axis
__('Hamster*0:acceleration.x').d

# Acceleration value on y-axis
__('Hamster*0:acceleration.y').d

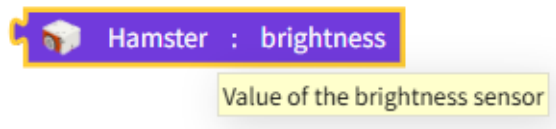
# Acceleration value on z-axis
__('Hamster*0:acceleration.z').d
```

Light Sensor Value

Returns the Hamster's light sensor value.

The value increases when the surroundings are bright, and decreases when it is dark.

The range is 0 to 600.



JavaScript Code

```
// Light sensor value
$('Hamster*0:light').d;
```

Python Code

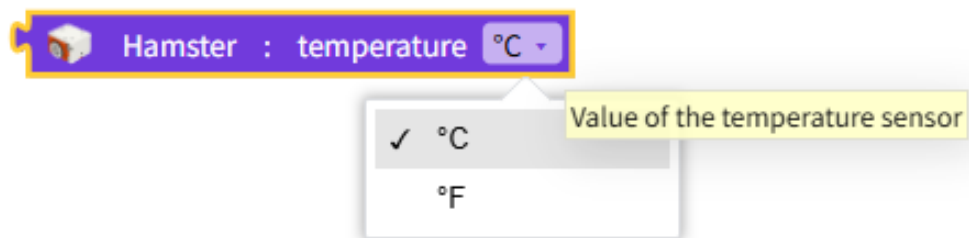
```
# Light sensor value
__('Hamster*0:light').d
```

Temperature Sensor Value

Returns the Hamster's temperature sensor value.

It represents the internal temperature of the robot, not the room temperature.

The range is -40.0 to +87.5 degrees(°C).



Dropdown Options and Input Values

Name	Type	Description	Range / Type
unit	Dropdown Option	Temperature unit	°C, °F

JavaScript Code

```
// Temperature in Celsius
__getTemperature($('Hamster*0:temperature').d, '°C');

// Temperature in Fahrenheit
__getTemperature($('Hamster*0:temperature').d, '°F')
```

Python Code

```
# Temperature in Celsius
__getTemperature(__('Hamster*0:temperature').d, '°C')

# Temperature in Fahrenheit
__getTemperature(__('Hamster*0:temperature').d, '°F')
```

Signal Strength Value

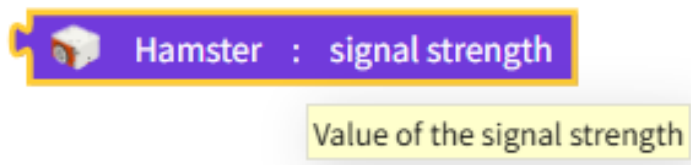
Returns the Hamster's signal strength value.

It indicates the signal strength between the robot and the dongle.

The value increases as the distance decreases.

For the stable operation, it is recommended to use within the range of -50dBm ~ -80dBm.

If the signal strength value is closer to 0, the signal is stronger. (e.g. -40 > -90)



JavaScript Code

```
// Signal strength value
$('Hamster*0:signal_strength').d;
```

Python Code

```
# Signal strength value
__('Hamster*0:signal_strength').d
```

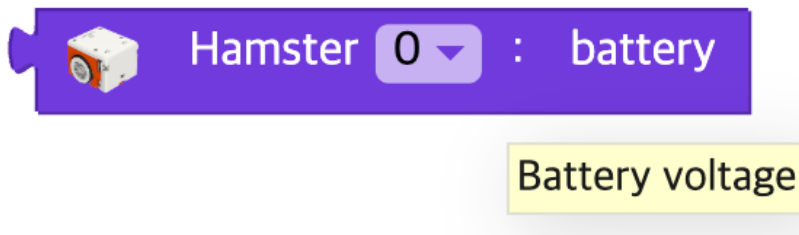
Battery Voltage

Returns the Hamster's battery voltage value.

The voltage is approximatedly 4.0V when the battery is fully charged.

It typically operates stably at 3.7V or higher.

Since operating errors may occur if the voltage drops below 3.5V, charging is strongly recommended.



JavaScript Code

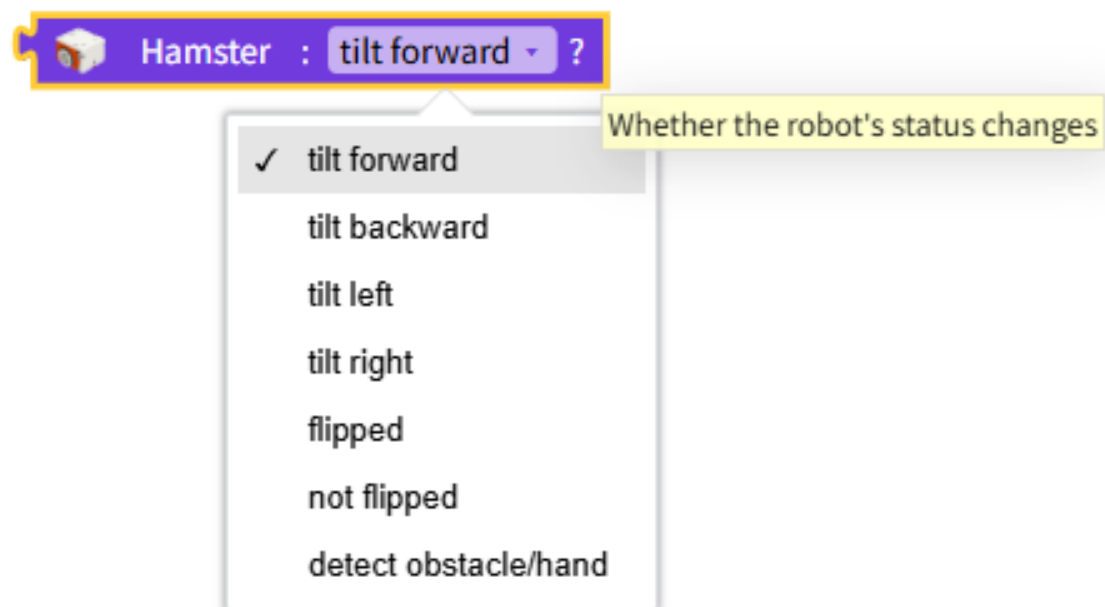
```
// Battery voltage value  
$('Hamster*0:battery.level').d;
```

Python Code

```
# Battery voltage value  
__('Hamster*0:battery.level').d
```

State Change Condition

Depending on the condition, returns whether the Hamster's state has changed as **True (1)** / **False (0)**.



Dropdown Options and Input Values

Name	Type	Description	Range / Type
condition	Dropdown Option	Pose / detect condition	tilt forward, tilt backward, tilt left, tilt right, flipped, not flipped, detected obstacle/hand

JavaScript Code

```
// Is Hamster tilted forward?
$('Hamster*0:acceleration.x').d > 5000;

// Is Hamster tilted backward?
$('Hamster*0:acceleration.x').d < -5000;

// Is Hamster tilted left?
$('Hamster*0:acceleration.y').d > 5000;

// Is Hamster tilted right?
$('Hamster*0:acceleration.y').d < -5000;

// Is Hamster flipped?
$('Hamster*0:acceleration.z').d > 0;

// Is Hamster not flipped?
$('Hamster*0:acceleration.z').d < -3000;

// Did Hamster detect obstacle/hand?
$('Hamster*0:proximity.left').d > 50 || $('Hamster*0:proximity.right').d > 50;
```

Python Code

```
# Is Hamster tilted forward?
__('Hamster*0:acceleration.x').d > 5000

# Is Hamster tilted backward?
__('Hamster*0:acceleration.x').d < -5000

# Is Hamster tilted left?
__('Hamster*0:acceleration.y').d > 5000

# Is Hamster tilted right?
__('Hamster*0:acceleration.y').d < -5000

# Is Hamster flipped over?
__('Hamster*0:acceleration.z').d > 0

# Is Hamster not flipped?
__('Hamster*0:acceleration.z').d < -3000

# Did Hamster detect obstacle/hand?
__('Hamster*0:proximity.left').d > 50 or __('Hamster*0:proximity.right').d > 50
```

Set IO Port Input Mode

Sets the input mode of the Hamster I/O ports.

You can set port A, port B, or both ports A and B.



Set the input mode of the IO port.

- ✓ analog input
- digital input
- servo output
- pwm output
- digital output

Dropdown Options and Input Values

Name	Type	Description	Range / Type
port	Dropdown Option	I/O port selection	A, B, A and B
mode	Dropdown Option	Port mode	analog input, digital input, servo output, pwm output, digital output

JavaScript Code

```
// Set Hamster port A to analog input
$('Hamster*0:io.a.mode').d = 0;

// Set Hamster port A to digital input
$('Hamster*0:io.a.mode').d = 1;

// Set Hamster port B to servo output
$('Hamster*0:io.b.mode').d = 8;

// Set Hamster port B to pwm output
$('Hamster*0:io.b.mode').d = 9;
```

```
// Set Hamster ports A and B to digital output
$('Hamster*0:io.a.mode').d = 10;
$('Hamster*0:io.b.mode').d = 10;
```

Python Code

```
# Set Hamster port A to analog input
__('Hamster*0:io.a.mode').d = 0

# Set Hamster port A to digital input
__('Hamster*0:io.a.mode').d = 1

# Set Hamster port B to servo output
__('Hamster*0:io.b.mode').d = 8

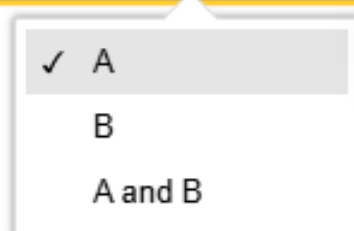
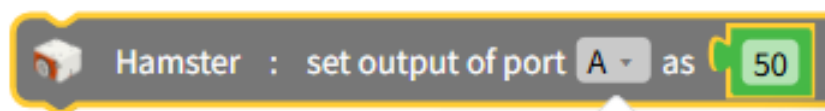
# Set Hamster port B to pwm output
__('Hamster*0:io.b.mode').d = 9

# Set Hamster ports A and B to digital output
__('Hamster*0:io.a.mode').d = 10
__('Hamster*0:io.b.mode').d = 10
```

Set IO Port Output Value

Sets the output value of the Hamster I/O ports.

You can set port A, port B, or both ports A and B.



Set the output value of the selected IO port.

Dropdown Options and Input Values

Name	Type	Description	Range / Type
port	Dropdown Option	I/O port selection	A, B, A and B
output	Input Value	Output value	0 ~ 180

JavaScript Code

```
// Set Hamster port A output to 50
$('Hamster*0:io.a.out').d = 50;

// Set Hamster port B output to 40
$('Hamster*0:io.b.out').d = 40;

// Set Hamster ports A and B output to 30
$('Hamster*0:io.a.out').d = 30;
$('Hamster*0:io.b.out').d = 30;
```

Python Code

```
# Set Hamster port A output to 50
__('Hamster*0:io.a.out').d = 50

# Set Hamster port B output to 40
__('Hamster*0:io.b.out').d = 40

# Set Hamster ports A and B output to 30
__('Hamster*0:io.a.out').d = 30
__('Hamster*0:io.b.out').d = 30
```

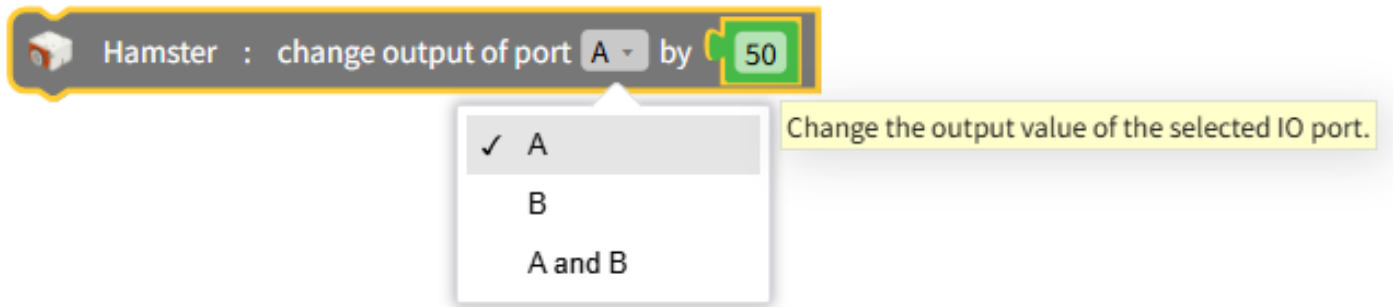
Change IO Port Output Value

Changes the output value of the Hamster I/O ports.

The new output value becomes the current port output value plus the input value.

The new output value is clamped to the range 0 ~ 180.

You can set port A, port B, or both ports A and B.



Dropdown Options and Input Values

Name	Type	Description	Range / Type
port	Dropdown Option	I/O port selection	A, B, A and B
value	Input Value	Delta value	Integer

JavaScript Code

```
// Change Hamster port A output by +10
$('Hamster*0:io.a.out').d = $('Hamster*0:io.a.out').d + 10;

// Change Hamster port B output by +20
$('Hamster*0:io.b.out').d = $('Hamster*0:io.b.out').d + 20;

// Change Hamster ports A and B output by +30
$('Hamster*0:io.a.out').d = $('Hamster*0:io.a.out').d + 30;
$('Hamster*0:io.b.out').d = $('Hamster*0:io.b.out').d + 30;
```

Python Code

```
# Change Hamster port A output by +10
__('Hamster*0:io.a.out').d = __('Hamster*0:io.a.out').d + 10

# Change Hamster port B output by +20
__('Hamster*0:io.b.out').d = __('Hamster*0:io.b.out').d + 20
```

```
# Change Hamster ports A and B output by +30
__('Hamster*0:io.a.out').d = __('Hamster*0:io.a.out').d + 30
__('Hamster*0:io.b.out').d = __('Hamster*0:io.b.out').d + 30
```

Open / Close Gripper

Opens or closes the Hamster's gripper.



Dropdown Options and Input Values

Name	Type	Description	Range / Type
toggle	Dropdown Option	Gripper toggle	open(1), close(2)

JavaScript Code

```
// Open Hamster gripper
$('Hamster*0:io.gripper').d = 1;

// Close Hamster gripper
$('Hamster*0:io.gripper').d = 2;
```

Python Code

```
# Open Hamster gripper
__('Hamster*0:io.gripper').d = 1

# Close Hamster gripper
__('Hamster*0:io.gripper').d = 2
```

Set Shooter Angle

Controls the Hamster by setting the shooter angle.

Angle range is 0 ~ 180.



Set the shooter angle. The angle ranges from 0 to 180.

Dropdown Options and Input Values

Name	Type	Description	Range / Type
angle	Input Value	Shooter angle	Integer 0 ~ 180

JavaScript Code

```
// Set Hamster shooter angle to 180
$('Hamster*0:io.shooter').d = 180;

// Set Hamster shooter angle to 0
$('Hamster*0:io.shooter').d = 0;
```

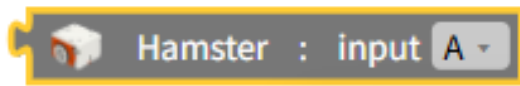
Python Code

```
# Set Hamster shooter angle to 180
__('Hamster*0:io.shooter').d = 180

# Set Hamster shooter angle to 0
__('Hamster*0:io.shooter').d = 0
```

IO Port Input Value

Returns the Hamster I/O port input value.



Input value of the IO port.

Dropdown Options and Input Values

Name	Type	Description	Range / Type
port	Dropdown Option	I/O port	A, B

JavaScript Code

```
// Hamster port A input value
$('Hamster*0:io.a.in').d;

// Hamster port B input value
$('Hamster*0:io.b.in').d;
```

Python Code

```
# Hamster port A input value
__('Hamster*0:io.a.in').d

# Hamster port B input value
__('Hamster*0:io.b.in').d
```